

Lab-8 Ex2 - Functions Warmup (II)

(Prelab*)

This is a PreLab exercise. Please do the PreLabs after each lecture and submit before you come to lab!

Printing out areas and circumferences of circles is boring! How about we draw the circles?

User-input numbers are ***bold and italic*** in the following sample runs.

Sample Run #1

Radius? **12**

[illegible]

Sample Run #2

Radius? **2**

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— ୧୧ —

Please implement the function pseudocode below to make it a reality! **Can you see how it works?** Please DO NOT TOUCH the main program.

```
function PrintCircle (returns nothing, takes in integer parameter 'radi
    declare integers i,j
    declare double centerX, centerY
    centerX = centerY = radius
    for i = 0 to radius * 2 - 1
        for j = 0 to radius * 2 - 1
            declare double posX = i + 0.5
            declare double posY = j + 0.5
            declare double distance
                = sqrt( pow(posX-centerX, 2) + pow(posY-centerY, 2) )
            if distance > radius print "-"
            otherwise, print "@"
        end for
        print new line character
    end for
end function
```